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Micro-optomechanical Sensor and Fabrication Method Thereof

Abstract

An example embodiment includes a micro-optomechanical sensor. The micro-optomechanical sensor includes a surface comprising a cavity, and a fluidic channel on the surface connected to the cavity. The fluidic channel is configured to transport a fluid sample to the cavity. The micro-optomechanical sensor also includes a cantilever on the surface being laterally extended inside the cavity. At least one side of the cantilever is configured to be in contact with the fluid sample inside the cavity. The micro-optomechanical sensor also includes an optical waveguide on the surface being extended to the cantilever. The optical waveguide is configured to couple out a light signal from the cantilever in order to optically transduce a mechanical vibration of the cantilever.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a non-provisional patent application claiming priority to European Patent Application No. 24160890.0, filed Mar. 1, 2024, the contents of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to opto-mechanical sensing devices, especially for bio-sensing based on mechanical vibration.

BACKGROUND

[0003] Generally, sensing a small vibration or motion can be useful in a variety of applications. It can also be used to characterize any biological system that may have a signature motion. Moreover, a fabrication technique to facilitate highly sensitive nano-motion sensors can be extremely useful for various fields. One such application may be to screen a drug or antibody. In this regard, the sensor can be used to detect a change in the living body in terms of a change in vibration due to the application of the drug or antibody. In some approaches, a cantilever arrangement with a micro-channel may be provided to translate the mass changes of the fluid traversing through the channel into changes in resonance frequency. However, the change in the vibration of the cantilever may be affected by the flow of the fluid through the channel formed in the cantilever.

SUMMARY

[0004] To address these and other issues, this disclosure provides a micro-optomechanical sensor and a fabrication method thereof for facilitating an improved bio-sensing scheme based on mechanical vibration, which is also suitable for massive parallelization.

[0005] In the disclosure, a micro-optomechanical sensor is provided. The sensor comprises a surface comprising a cavity, and a fluidic channel on the surface connected to the cavity, wherein the fluidic channel is configured to transport a fluid sample to the cavity. The sensor further comprises a cantilever on the surface being laterally extended inside the cavity, wherein at least one side of the cantilever is configured to be in contact with the fluid sample inside the cavity.

[0006] Moreover, the sensor comprises an optical waveguide on the surface being extended to the cantilever, wherein the optical waveguide is configured to couple out a light signal from the cantilever in order to optically transduce a mechanical vibration of the cantilever.

[0007] Therefore, the cavity, especially the micro-cavity, may be used as a reservoir for the fluid sample and hence the vibration of the cantilever may not be affected by the flow of the fluid sample, which may improve the sensing accuracy. Moreover, by means of the integrated optical waveguide, a change in the vibration of the cantilever can be optically translated to identify an interaction or to map multiple interactions of the cantilever with the sample fluid within the cavity. For example, the fluid sample is a biological sample comprising at least one biological cell to be classified and/or identified.

[0008] In some embodiments, the at least one side of the cantilever comprises a coating of at least one antibody and/or at least one antigen and/or at least one protein and/or at least one aptamer corresponding to at least one biological cell in the fluid sample, e.g., the at least one biological cell of the biological sample.

[0009] In some embodiments, the mechanical vibration of the cantilever corresponds to a natural resonance frequency. In other embodiments, the mechanical vibration of the cantilever corresponds to a resonance frequency higher or lower than the natural resonance frequency due to an interaction of the at least one biological cell with the coating. In this way, cell-specific signatures can be effectively identified and/or classified by detecting the changes in the resonance frequency.

[0010] In some embodiments, the surface further comprises an inlet connected to the fluidic

channel, wherein the inlet is configured to receive the fluid sample and further to transport the fluid sample to the fluidic channel.

[0011] In some embodiments, the cantilever is arranged on the surface in a non-overlapping manner with respect to the fluidic channel on the surface. Additionally or alternatively, the cantilever is arranged on the surface in a non-overlapping manner with respect to the inlet on the surface. In other embodiments, the cantilever is arranged on the surface in a non-overlapping manner with respect to both the inlet and the fluidic channel on the surface. In this way, the flow of the fluid sample may not be affected by the motion or vibration of the cantilever, which may further improve the sensing accuracy.

[0012] In some embodiments, the optical waveguide is a dielectric waveguide, such as a silicon-based waveguide. In this way, a low transmission loss and a good light confinement can be achieved, especially for the light traversing along the surface as well as along the cantilever.

[0013] In some embodiments, the surface is a processed silicon-on-insulator, SOI, wafer. In this way, a high yield can be achieved, especially for realizing a large number of parallel sensors in a highly dense manner.

[0014] According to a second aspect of the disclosure, a sensor system is provided. The sensor system comprises at least one micro-optomechanical sensor such as that described above. The sensor system further comprises at least one light source optically coupled to the optical waveguide of the at least one micro-optomechanical sensor, the optical waveguides of the respective plurality of micro-optomechanical sensors, wherein the at least one light source is configured to generate the light signal for the optical waveguide of the at least one micro-optomechanical sensor, for each of the optical waveguides of the respective plurality of micro-optomechanical sensors.

[0015] Moreover, the sensor system comprises a processing device, especially arranged in relation to the cantilever of the at least one micro-optomechanical sensor, to the cantilevers of the respective plurality of micro-optomechanical sensors, the processing device being optically coupled to the at least one micro-optomechanical sensor, to the plurality of micro-optomechanical sensors, wherein the processing device is configured to receive the light signal coupled out from the optical waveguide of the at least one micro-optomechanical sensor, from each of the optical waveguides of the respective plurality of micro-optomechanical sensors, and further to optically transduce the mechanical vibration of the cantilever of the at least one micro-optomechanical sensor, the mechanical vibration of the cantilevers of the respective plurality of micro-optomechanical sensors.

[0016] In this regard, for the at least one micro-optomechanical sensor, for each of the plurality of the micro-optomechanical sensors, the processing device is further configured to process the light signal coupled out from the optical waveguide to detect a change in the resonance frequency of the mechanical vibration of the cantilever due to the interaction of the at least one biological cell with the coating, and to classify and/or identify the at least one biological cell in the fluid sample based on the detected change in the resonance frequency.

[0017] In this way, the parallelization of the micro-optomechanical sensors can be achieved, which may reduce the screening time, e.g., to classify and/or identify one or more cell-specific signatures simultaneously or sequentially.

[0018] In the disclosure, a method is provided for fabricating a micro-optomechanical sensor on a SOI wafer comprising a dielectric layer in between a top silicon layer and a bottom silicon layer. The method comprises a step of defining a length of a cantilever and a length of a cavity on the SOI wafer. In addition, the method comprises a step of processing an optical waveguide on the top silicon layer, the optical waveguide being extended over the length of the cantilever.

[0019] Furthermore, the method comprises a step of processing a first etched groove at an edge of the length of the cantilever within the length of the cavity by etching the top silicon wafer, the dielectric layer, and partially the bottom silicon layer from the top silicon layer side.

[0020] In addition, the method comprises a step of processing a second etched groove

corresponding to the length of the cavity by etching the bottom silicon layer and the dielectric layer from the bottom silicon layer side, thereby forming the cavity in the second silicon layer and the cantilever on the first silicon layer laterally extended inside the cavity.

[0021] The method further comprises a step of bonding a substrate on the bottom silicon layer at least encompassing the cavity from the bottom silicon layer side. Moreover, the method comprises a step of processing a fluidic channel on the top silicon layer from the top silicon layer side.

[0022] In this way, a miniaturization of the micro-optomechanical sensor can be achieved, which may reduce the cost and size, and may allow for massive parallelization.

[0023] In some embodiments, the processing of the optical waveguide comprises a step of forming optical waveguide components on the top silicon layer from the top silicon layer side. Alternatively, the processing of the optical waveguide comprises a step of bonding an additional layer on the top silicon layer and forming optical waveguide components on the additional layer from the top silicon layer side. In this way, the optical waveguide can be processed on the top silicon layer in a flexible manner based on, e.g., the thickness of the sensor, the wavelength of the light, and so on.

[0024] In some embodiments, the processing of the first etched groove and/or the processing of the second etched groove further comprise a step of using the dielectric layer in between the top silicon layer and the bottom silicon layer as a etch stop layer. In this way, the processing of the grooves can be effectively and accurately performed.

[0025] In some embodiments, the bonding of the substrate on the bottom silicon layer comprises a step of bonding the substrate on the bottom silicon layer via oxide fusion between the substrate and the bottom silicon layer. In some embodiments, the bonding of the substrate on the bottom silicon layer comprises a step of bonding the substrate on the bottom silicon layer via an adhesive layer in between the substrate and the bottom silicon layer. In some embodiments, for example, the cavity in the bottom silicon layer can be effectively sealed in a flexible manner.

[0026] In some embodiments, the processing of the fluidic channel comprises the steps of laminating a dry-film-resist, DFR, on the top silicon layer, patterning the fluidic channel via lithography on the DFR, and hard-baking the patterned DFR. In this way, the fluidic channel can be processed in a simplified manner, whereby additionally protecting the surface and its components by means of the laminated DFR.

[0027] In a first aspect, the disclosure describes a micro-optomechanical sensor. The micro-optomechanical sensor includes a surface comprising a cavity, and a fluidic channel on the surface connected to the cavity. The fluidic channel is configured to transport a fluid sample to the cavity. The micro-optomechanical sensor also includes a cantilever on the surface being laterally extended inside the cavity. At least one side of the cantilever is configured to be in contact with the fluid sample inside the cavity. The micro-optomechanical sensor also includes an optical waveguide on the surface being extended to the cantilever. The optical waveguide is configured to couple out a light signal from the cantilever in order to optically transduce a mechanical vibration of the cantilever.

[0028] In a second aspect, the disclosure describes a sensor system. The sensor system includes at least one micro-optomechanical sensor in accordance with the first aspect. The sensor system also includes at least one light source optically coupled to the optical waveguide of the at least one micro-optomechanical sensor. The at least one light source is configured to generate the light signal for one or more optical waveguides of the at least one micro-optomechanical sensor. The sensor system also includes a processing device optically coupled to the at least one micro-optomechanical sensor. The processing device is configured to receive the light signal coupled out from each of the optical waveguides of the at least one micro-optomechanical sensor and further to optically transduce the mechanical vibration of one or more cantilevers of the at least one micro-optomechanical sensor.

[0029] In a third aspect, the disclosure describes a method for fabricating a micro-optomechanical sensor on a silicon-on-insulator (SOI) wafer comprising a dielectric layer in between a top silicon

layer and a bottom silicon layer. The method includes defining a length of a cantilever and a length of a cavity on the SOI wafer. The method also includes processing an optical waveguide on the top silicon layer. The optical waveguide is extended over the length of the cantilever. The method also includes processing a first etched groove at an edge of the length of the cantilever within the length of the cavity by etching the top silicon layer, the dielectric layer, and partially the bottom silicon layer from the top silicon layer side. The method also includes processing a second etched groove corresponding to the length of the cavity by etching the bottom silicon layer and the dielectric layer from the bottom silicon layer side, thereby forming the cavity in the bottom silicon layer and the cantilever on the top silicon layer laterally extended inside the cavity. The method also includes bonding a substrate on the bottom silicon layer at least encompassing the cavity from the bottom silicon layer side. The method also includes processing a fluidic channel on the top silicon layer from the top silicon layer side.

[0030] The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the figures and the following detailed description.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0031] FIG. 1 shows a micro-optomechanical sensor according to example embodiments.

[0032] FIG. 2 shows a sensor system according to an example embodiments.

[0033] FIG. 3A shows a silicon-on-insulator (SOI) wafer according to example embodiments.

[0034] FIG. 3B shows a process for forming an optical waveguide on the SOI wafer according to example embodiments.

[0035] FIG. 3C shows an alternative process for forming the optical waveguide on the SOI wafer according to example embodiments.

[0036] FIG. 4A shows a process for defining the in-plane geometry of the micro-optomechanical sensor according to example embodiments.

[0037] FIG. 4B shows an alternative process for defining the in-plane geometry of the micro-optomechanical sensor according to example embodiments.

[0038] FIG. 5A shows a process for forming the cavity and the cantilever according to example embodiments.

[0039] FIG. 5B shows an alternative process for forming the cavity and the cantilever according to example embodiments.

[0040] FIG. 6A shows a process for sealing the cavity according to example embodiments.

[0041] FIG. 6B shows an alternative process for sealing the cavity according to example embodiments.

[0042] FIG. 7A shows a process for forming the fluidic channel according to example embodiments.

[0043] FIG. 7B shows an alternative process for forming the fluidic channel according to example embodiments.

[0044] FIG. 8 shows an exemplary flow diagram of the method according to example embodiments.

DETAILED DESCRIPTION

[0045] Example methods and systems are described herein. Any example embodiment or feature described herein is not necessarily to be construed as preferred or advantageous over other embodiments or features. The example embodiments described herein are not meant to be limiting. It will be readily understood that certain aspects of the disclosed systems and methods can be

arranged and combined in a wide variety of different configurations, all of which are contemplated herein.

[0046] Furthermore, the particular arrangements shown in the figures should not be viewed as limiting. It should be understood that other embodiments might include more or less of each element shown in a given figure. In addition, some of the illustrated elements may be combined or omitted. Similarly, an example embodiment may include elements that are not illustrated in the figures.

[0047] It is important to note that, in the description as well as in the claims, the word “comprising” does not exclude other elements or steps and the indefinite article “a” or “an” does not exclude a plurality. A single element or other unit may fulfill the functions of several entities or items recited in the claims.

[0048] It should be understood that the term “and/or” used in the specification of this application refers to any combination and all possible combinations of one or more associated listed items, and includes these combinations.

[0049] It should also be understood that the word “connected” implies that the elements may be directly connected together or may be coupled through one or more intervening elements. Moreover, the disclosure with regard to any of the aspects is also relevant with regard to the other aspects of the disclosure.

[0050] Reference will now be made in detail to the embodiments of the present disclosure, examples of which are illustrated in the accompanying drawings. However, the following embodiments of the present disclosure may be variously modified and the range of the present disclosure is not limited by the following embodiments.

I. Example Micro-Optomechanical Sensors

[0051] In FIG. 1, an example embodiment of the micro-optomechanical sensor **100** according to an aspect of the disclosure is illustrated. The sensor **100** may include a surface **101**, which may itself include a micro-cavity or a cavity **102**. The surface **101** may further include a micro-fluidic channel or a fluidic channel **103** connected to the cavity **102**. Moreover, the surface **101** may include a micro-inlet or an inlet **104** connected to the fluidic channel **103**.

[0052] In this regard, the inlet **104** may receive a biological sample fluid or a fluid sample and may transport the fluid sample to the fluidic channel **103**, where the fluidic channel **103** may further transport the fluid sample to the cavity **102**, which may act as a reservoir for the fluid sample.

[0053] In addition, the surface **101** may include a micro-cantilever or a cantilever **105** being laterally extended inside the cavity **102** such that one side, e.g., the bottom side, of the cantilever **105** may be in contact with the fluid sample inside the cavity **102**. Furthermore, the cantilever **105** may be arranged on the surface **101** such that the cantilever **105** may not overlap with the inlet **104** as well as the fluidic sample **103**.

[0054] In this regard, the bottom side of the cantilever **105** may be coated, e.g., the area or the portion of the bottom side of the cantilever **105** being in contact with the fluid sample, with one or more antibodies, or with one or more antigens, or with one or more proteins, or with one or more aptamers, or any combination thereof. Accordingly, the natural resonance frequency of the vibration of the cantilever **105** may be changed due to the interactions of the biological cells in the fluid sample with the coating.

[0055] The surface **101** may also include a micro-optical waveguide or an optical waveguide **106**. The optical waveguide **106** may include an input port **107** for receiving or coupling-in a light signal, e.g., from an external or an internal light source (not shown). The optical waveguide **106** may be extended to the cantilever **105**, for example, to the top side of the cantilever **105** opposite to the coated area of the cantilever **105**.

[0056] For example, especially at the top side of the cantilever **105**, the optical

[0057] waveguide **106** may comprise an output port (not shown) so that the optical waveguide **106** may couple out the light signal from the top side of the cantilever **105** via the output port in order

to optically translate the mechanical vibration of the cantilever **105**.

II. Example Sensor Systems

[0058] In FIG. **2**, an example embodiment of the sensor system **200** according to an aspect of the disclosure is illustrated. The sensor system **200** may include a first micro-optomechanical sensor **100a** and a second micro-optomechanical sensor **100b** being operated in parallel. However, in some embodiments, the sensor system **200** may include more than two micro-optomechanical sensors **100a**, **100b** that can be operated in parallel.

[0059] For example, the first sensor **100a** may correspond to the micro-optomechanical sensor **100** of FIG. **1**. In this regard, the surface **101a**, the cavity **102a**, the fluidic channel **103a**, the inlet **104a**, the cantilever **105a**, the optical waveguide **106a**, and the light input port **107a** may correspond to the surface **101**, the cavity **102**, the fluidic channel **103**, the inlet **104**, the cantilever **105**, the optical waveguide **106**, and the light input port **107** of the micro-optomechanical sensor **100**, respectively.

[0060] The second sensor **100b** may also correspond to the micro-optomechanical sensor **100** of FIG. **1**. In this regard, the surface **101b**, the cavity **102b**, the fluidic channel **103b**, the inlet **104b**, the cantilever **105b**, the optical waveguide **106b**, and the light input port **107b** may correspond to the surface **101**, the cavity **102**, the fluidic channel **103**, the inlet **104**, the cantilever **105**, the optical waveguide **106**, and the light input port **107** of the micro-optomechanical sensor **100**, respectively.

[0061] In some embodiments, the fluid sample of the first sensor **100a** and the fluid sample of the second sensor **100b** may be different fluid samples, e.g., containing different types of biological cells. Alternatively, the fluid sample of the first sensor **100a** and the fluid sample of the second sensor **100b** may be similar fluid samples, e.g., containing similar types of biological cells.

[0062] In some embodiments, the coating of the cantilever **105a** of the first sensor **100a** and the coating of the cantilever **105b** of the second sensor **100b** may be different coatings, e.g., containing different types of antibodies and/or antigens and/or proteins and/or aptamers. Alternatively, the coating of the cantilever **105a** of the first sensor **100a** and the coating of the cantilever **105b** of the second sensor **100b** may be similar coatings, e.g., containing similar types of antibodies and/or antigens and/or proteins and/or aptamers.

[0063] The sensor system **200** may further include a light source **201**, e.g., a laser light source such as a laser diode. The light source **200** may be optically connected to the light input port **107a** of the first sensor **100a** via a first optical path **202** such that the light source **201** may provide a first light signal **204** for the optical waveguide **106a** of the first sensor **100a** via the first optical path **202**.

[0064] In addition, the light source **200** may be optically connected to the light input port **107b** of the second sensor **100b** via a second optical path **203** such that the light source **201** may provide a second light signal **205** for the optical waveguide **106b** of the second sensor **100b** via the second optical path **203**.

[0065] In some embodiments, the first light signal **204** and the second light signal **205** may be light signals with similar electrical and/or optical characteristics, e.g., comprising identical or similar wavelengths. Alternatively, the first light signal **204** and the second light signal **205** may be light signals with different electrical and/or optical characteristics, e.g., comprising different wavelengths.

[0066] In this regard, the first optical path **202** and/or the second optical path **203** may include different electrical and/or optical modulation circuitry to provide light signals with different characteristics, e.g., wavelengths. Additionally or alternatively, the optical waveguide **106a** of the first sensor **100a** and/or the optical waveguide **106b** of the second sensor **100b** may include different electrical and/or optical modulation circuitry to provide light signals with different characteristics, e.g., wavelengths.

[0067] In some embodiments, the sensor system **200** may include a processing device **206** that is optically coupled to one or more sensors **100a**, **100b**. It may be noted that the term “optically coupled” is referred to herein as communicating optical signals with or without waveguide/optical fiber from the sensors **100a**, **100b** to the processing device **206**. In some embodiments, the

processing device **206** may include one or more light detectors configured to receive the light signals **204**, **205**. Additionally, the processing device **206** may include an imager or an imaging unit. The processing device **206** may be arranged in relation to the cantilever **105a** of the first sensor **100a** and the cantilever **105b** of the second sensor **100b** such that the processing device **206** may receive, especially in parallel, the first light signal **204** from the output port of the optical waveguide **106a** at the top side of the cantilever **105a** of the first sensor **100a** and the second light signal **205** from the output port of the optical waveguide **106b** at the top side of the cantilever **105b** of the second sensor **100b**.

[0068] In this regard, the processing device **206** may process the first light signal **204** and may detect a change in the resonance frequency of the mechanical vibration of the cantilever **105a** of the first sensor **100a**, e.g., due to the interaction of the first fluid sample with the coating, and may classify and/or identify one or more cells or the signature of the cells of the first fluid sample based on the detected change in the resonance frequency.

[0069] Similarly, the processing device **206** may process the second light signal **205** and may detect a change in the resonance frequency of the mechanical vibration of the cantilever **105b** of the second sensor **100b**, e.g., due to the interaction of the second fluid sample with the coating, and may classify and/or identify one or more cells or the signature of the cells of the second fluid sample based on the detected change in the resonance frequency.

III. Example Processes

[0070] In FIG. 3A, an example SOI wafer **300** according to an aspect of the disclosure is illustrated. The SOI wafer **300** may include a top silicon layer **301**, a bottom silicon layer **303**, and a dielectric layer **302**, e.g., an oxide-based dielectric such as SiO₂, in-between the top silicon layer **301** and the bottom silicon layer **303**.

[0071] In some embodiments, the bottom silicon layer **303** (or the “SOI substrate”) may have a thickness of about 1 micrometer to 300 micrometers. The dielectric layer **302** (or the “buried oxide layer”) may have a thickness of about 2 micrometers to 20 micrometers. The top silicon layer **301** (or the “device layer”) may have a thickness of about 1 micrometer to 30 micrometers.

[0072] In FIG. 3B, an example process for forming the optical waveguide **304a** on the SOI wafer **300** according to an aspect of the disclosure is illustrated. For example, a length of the cantilever may be defined on the top silicon layer **301**. Afterwards, the optical waveguide **304a** may be processed on the top silicon layer **301** from the top silicon layer side (front-end processing) such that the optical waveguide **304a** may extend along the defined length of the cantilever. In this regard, the top silicon layer **301** may be patterned to make the waveguide components in a suitable manner during the front-end processing.

[0073] In FIG. 3C, an example process for forming the optical waveguide **304b** on the SOI wafer **300** according to an aspect of the disclosure is illustrated. For example, an additional dielectric layer **305** may be added on the top silicon layer **301**, e.g., via fusion or dielectric bonding. Afterwards, the length of the cantilever may be defined on the additional layer **305**.

[0074] Furthermore, the optical waveguide **304b** may be processed on the additional layer **305** from the top silicon layer side (front-end processing) such that the optical waveguide **304b** may extend along the defined length of the cantilever. In this regard, the additional layer **305** may be patterned to make the waveguide components in a suitable manner during the front-end processing.

[0075] The additional layer **305** may have a thickness of about 1.5 micrometers. The optical waveguide **304b**, especially the waveguide components, may be realized as planar waveguide components, e.g., SiN waveguide components.

[0076] In FIG. 4A, an example process for defining the in-plane geometry of the micro-optomechanical sensor according to an aspect of the disclosure is illustrated. In this regard, a groove **401a** may be formed in the front-end processing, especially at an edge of the defined length of the cantilever, by etching the top silicon layer **301**, especially the processed top silicon layer as shown in FIG. 3A, the dielectric layer **302**, and partially the bottom silicon layer **303**.

[0077] In some embodiments, the dielectric layer **302** may be used as the stopping layer for the lithography and the subsequent etching to form the groove **401a**. In this regard, the groove **401a** may define the in-plane geometry of the sensor, especially of the cantilever of the sensor.

[0078] In FIG. **4B**, another example process for defining the in-plane geometry of the micro-optomechanical sensor according to an aspect of the disclosure is illustrated. In this regard, a groove **401b** may be formed in the front-end processing, especially at an edge of the defined length of the cantilever, by etching the additional layer **305**, especially the processed additional layer as shown in FIG. **3B**, the top silicon layer **301**, the dielectric layer **302**, and partially the bottom silicon layer **303**.

[0079] In some embodiments, the dielectric layer **302** may be used as the stopping layer for the lithography and the subsequent etching to form the groove **401b**. In this regard, the groove **401b** may define the in-plane geometry of the sensor, especially of the cantilever of the sensor.

[0080] In FIG. **5A**, an example process for forming a cavity and a cantilever according to an aspect of the disclosure is illustrated. As shown, a length of the cavity may be defined on the bottom silicon layer **303**. Afterwards, a groove **501a** may be formed from the bottom silicon layer side (back-end processing) by etching the bottom silicon layer **303** and the dielectric layer **302**.

[0081] In some embodiments, the dielectric layer **302** may be used as the stopping layer for the lithography and the subsequent etching to form the groove **501a**. In this regard, the groove **501a** may define the in-plane geometry of the cavity and may further release the cantilever **502a** with the optical waveguide **304a**, especially by conjoining with the groove **401a**.

[0082] In FIG. **5B**, another example process for forming the cavity and the cantilever according to an aspect of the disclosure is illustrated. As shown, the length of the cavity may be defined on the bottom silicon layer **303**. Afterwards, a groove **501b** may be formed from the bottom silicon layer side (back-end processing) by etching the bottom silicon layer **303** and the dielectric layer **302**.

[0083] In some embodiments, the dielectric layer **302** may be used as the stopping layer for the lithography and the subsequent etching to form the groove **501b**. In this regard, the groove **501b** may define the in-plane geometry of the cavity and may further release the cantilever **502b** with the additional layer **305** comprising the optical waveguide **304b**, especially by conjoining with the groove **401b**.

[0084] In FIG. **6A**, an example process for sealing the cavity according to an aspect of the disclosure is illustrated. As shown, a substrate **601** may be bonded to the bottom silicon layer **303**, for instance to the processed bottom silicon layer as shown in FIG. **5A**, in the back-end processing such that the substrate **601** may encompass at least the groove **501a**. As such, the substrate **601** may completely seal the groove **501a** to form the cavity in the bottom silicon layer **303**.

[0085] Accordingly, the sealing of the groove **501a** may result in the cavity in the bottom silicon layer **303**, which may surround the cantilever **502a** realized on the first silicon layer **301** with the optical waveguide **304a**.

[0086] In FIG. **6B**, another example process for sealing the cavity according to an aspect of the disclosure is illustrated. For example, the substrate **601** may be bonded to the bottom silicon layer **303**, for instance to the processed bottom silicon layer as shown in FIG. **5B**, in the back-end processing such that the substrate **601** may encompass at least the groove **501b**. As such, the substrate **601** may completely seal the groove **501b** to form the cavity in the bottom silicon layer **303**.

[0087] Accordingly, the sealing of the groove **501b** may result in the cavity in the bottom silicon layer **303**, which may surround the cantilever **502b** realized on the first silicon layer **301** with the additional layer **305** comprising the optical waveguide **304b**.

[0088] Regarding the substrate **601** of FIGS. **6A** and **6B**, the substrate **601** may be a silicon substrate or a glass substrate in some embodiments. For instance, the substrate **601** may be bonded to the bottom silicon layer **303** via oxide fusion between the substrate **601** and the bottom silicon layer **303**. Alternatively, the substrate **601** may be bonded to the bottom silicon layer **303** via an

adhesive layer **602** in between the substrate **601** and the bottom silicon layer **303**. The adhesive layer **602** may be a layer of polyimide-based adhesives or a layer of acrylic-based adhesives.

[0089] In FIG. 7A, an example process for forming the fluidic channel **701a** according to an aspect of the disclosure is illustrated. For example, the fluidic channel **701a** may be formed on the top silicon layer **301**, for instance on the processed top silicon layer comprising the cantilever **502a** with the optical waveguide **304a**, in the front-end processing.

[0090] In some embodiments, the fluidic channel **701a** may be formed in the front-end processing by laminating a dry-film-resist (DFR) on the processed top silicon layer comprising the cantilever **502a** with the optical waveguide **304a**. Afterwards, the DFR may be patterned via lithography such that a fluidic connection with the cavity can be formed, and the patterned DFR may be subsequently hard-baked in order to fix the patterns. This may result in the micro-optomechanical sensor **700a**.

[0091] It is to be noted that the fluidic channel **701a** may be alternatively processed after the process for defining the in-plane geometry of the micro-optomechanical sensor, especially after the processing of the groove **401a** to define the in-plane geometry of the cantilever, as shown in FIG. 4A above.

[0092] In FIG. 7B, another example process for forming the fluidic channel **701b** according to an aspect of the disclosure is illustrated. For example, the fluidic channel **701b** may be formed on the additional layer **305**, for instance on the processed additional layer comprising the optical waveguide **304b**, in the front-end processing.

[0093] In some embodiments, the fluidic channel **701b** may be formed in the front-end processing by laminating a dry-film-resist (DFR) on the processed additional layer comprising the optical waveguide **304b**. Afterwards, the DFR may be patterned via lithography such that a fluidic connection with the cavity can be formed, and the patterned DFR may be subsequently hard-baked in order to fix the patterns. This may result in the micro-optomechanical sensor **700b**.

[0094] It is to be noted that the fluidic channel **701b** may be alternatively processed after the process for defining the in-plane geometry of the micro-optomechanical sensor, especially after the processing of the groove **401b** to define the in-plane geometry of the cantilever, as shown in FIG. 4B above.

[0095] In FIG. 8, an exemplary flow diagram of the method **800** according to the third aspect of the invention is illustrated. In a first step **801**, a length of a cantilever and a length of a cavity are defined on a SOI wafer comprising a dielectric layer in between a top silicon layer and a bottom silicon layer. In a second step **802**, an optical waveguide is processed on the top silicon layer such that the optical waveguide extends over the length of the cantilever.

[0096] In a third step **803**, a first etched groove is processed at an edge of the length of the cantilever within the length of the cavity by etching the top silicon wafer, the dielectric layer, and partially the bottom silicon layer from the top silicon layer side.

[0097] In a fourth step **804**, a second etched groove corresponding to the length of the cavity is processed by etching the bottom silicon layer and the dielectric layer from the bottom silicon layer side in order to form the cavity in the second silicon layer and the cantilever on the first silicon layer laterally extended inside the cavity.

[0098] In a fifth step **805**, a substrate is bonded on the bottom silicon layer in order to encompass at least the cavity from the bottom silicon layer side. In a sixth step **806**, a fluidic channel on the top silicon layer is processed from the top silicon layer side.

[0099] Although the invention has been illustrated and described with respect to one or more implementations, equivalent alterations and modifications will occur to others skilled in the art upon the reading and understanding of this specification and the annexed drawings. In addition, while a particular feature of the invention may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application.

IV. Enumerated Example Embodiments

[0100] Embodiments of the present disclosure may thus relate to one of the enumerated example embodiments (EEEs) listed below.

[0101] EEE 1 is a micro-optomechanical sensor comprising: [0102] a surface comprising a cavity; [0103] a fluidic channel on the surface connected to the cavity, wherein the fluidic channel is configured to transport a fluid sample to the cavity; [0104] a cantilever on the surface being laterally extended inside the cavity, wherein at least one side of the cantilever is configured to be in contact with the fluid sample inside the cavity; and [0105] an optical waveguide on the surface being extended to the cantilever, wherein the optical waveguide is configured to couple out a light signal from the cantilever in order to optically transduce a mechanical vibration of the cantilever. [0106] EEE 2 is the micro-optomechanical sensor according to EEE 1, wherein the at least one side of the cantilever comprises a coating of at least one antibody and/or at least one antigen and/or at least one protein and/or at least one aptamer, corresponding to at least one biological cell in the fluid sample.

[0107] EEE 3 is the micro-optomechanical sensor according to EEE 2, wherein the mechanical vibration of the cantilever corresponds to a natural resonance frequency.

[0108] EEE 4 is the micro-optomechanical sensor according to EEE 2, wherein the mechanical vibration of the cantilever corresponds to a resonance frequency higher or lower than the natural resonance frequency due to an interaction of the at least one biological cell with the coating.

[0109] EEE 5 is the micro-optomechanical sensor according to EEE 1, wherein the surface further comprises an inlet connected to the fluidic channel, wherein the inlet is configured to receive the fluid sample and further to transport the fluid sample to the fluidic channel.

[0110] EEE 6 is the micro-optomechanical sensor according to EEE 1, wherein the cantilever is arranged on the surface in a non-overlapping manner with respect to the fluidic channel on the surface.

[0111] EEE 7 is the micro-optomechanical sensor according to EEE 1, wherein the [0112] optical waveguide is a dielectric waveguide.

[0113] EEE 8 is the micro-optomechanical sensor according to EEE 7, wherein the dielectric waveguide is a silicon-based waveguide.

[0114] EEE 9 is the micro-optomechanical sensor according to EEE 1, wherein the surface is a processed silicon-on-insulator (SOI) wafer.

[0115] EEE 10 is a sensor system comprising: [0116] at least one micro-optomechanical sensor comprising: [0117] a surface comprising a cavity, [0118] a fluidic channel on the surface connected to the cavity, wherein the fluidic channel is configured to transport a fluid sample to the cavity, [0119] a cantilever on the surface being laterally extended inside the cavity, wherein at least one side of the cantilever is configured to be in contact with the fluid sample inside the cavity, and [0120] an optical waveguide on the surface being extended to the cantilever, wherein the optical waveguide is configured to couple out a light signal from the cantilever in order to optically transduce a mechanical vibration of the cantilever; [0121] at least one light source optically coupled to the optical waveguide of the at least one micro-optomechanical sensor, wherein the at least one light source is configured to generate the light signal for one or more optical waveguides of the at least one micro-optomechanical sensor; and [0122] a processing device optically coupled to the at least one micro-optomechanical sensor, wherein the processing device is configured to receive the light signal coupled out from each of the optical waveguides of the at least one micro-optomechanical sensor and further to optically transduce the mechanical vibration of one or more cantilevers of the at least one micro-optomechanical sensor.

[0123] EEE 11 is the sensor system of EEE 10, wherein the at least one side of the cantilever comprises a coating of at least one antibody and/or at least one antigen and/or at least one protein and/or at least one aptamer, corresponding to at least one biological cell in the fluid sample.

[0124] EEE 12 is the sensor system according to EEE 11, wherein the processing device is further

configured to: [0125] process the light signal coupled out from the optical waveguide to detect a change in the resonance frequency of the mechanical vibration of the cantilever due to the interaction of the at least one biological cell with the coating, and [0126] identify at least one biological cell in the fluid sample based on the detected change in the resonance frequency. [0127] EEE 13 is a method for fabricating a micro-optomechanical sensor on a silicon-on-insulator (SOI) wafer comprising a dielectric layer in between a top silicon layer and a bottom silicon layer, the method comprising: [0128] defining a length of a cantilever and a length of a cavity on the SOI wafer; [0129] processing an optical waveguide on the top silicon layer, wherein the optical waveguide is extended over the length of the cantilever; [0130] processing a first etched groove at an edge of the length of the cantilever within the length of the cavity by etching the top silicon layer, the dielectric layer, and partially the bottom silicon layer from the top silicon layer side; [0131] processing a second etched groove corresponding to the length of the cavity by etching the bottom silicon layer and the dielectric layer from the bottom silicon layer side, thereby forming the cavity in the bottom silicon layer and the cantilever on the top silicon layer laterally extended inside the cavity; [0132] bonding a substrate on the bottom silicon layer at least encompassing the cavity from the bottom silicon layer side; and [0133] processing a fluidic channel on the top silicon layer from the top silicon layer side.

[0134] EEE 14 is the method according to EEE 13, wherein the processing of the optical waveguide comprises: [0135] forming optical waveguide components on the top silicon layer from the top silicon layer side.

[0136] EEE 15 is the method according to EEE 13, wherein the processing of the optical waveguide comprises: [0137] bonding an additional layer on the top silicon layer and forming optical waveguide components on the additional layer from the top silicon layer side.

[0138] EEE 16 is the method according to EEE 13, wherein the processing of the first etched groove further comprises using the dielectric layer in between the top silicon layer and the bottom silicon layer as a etch stop layer.

[0139] EEE 17 is the method according to EEE 13, wherein the processing of the second etched groove further comprises using the dielectric layer in between the top silicon layer and the bottom silicon layer as a etch stop layer.

[0140] EEE 18 is the method according to EEE 13,

[0141] wherein the bonding of the substrate on the bottom silicon layer comprises: bonding the substrate on the bottom silicon layer via oxide fusion between the substrate and the bottom silicon layer.

[0142] EEE 19 is the method according to EEE 13, wherein the bonding of the substrate on the bottom silicon layer comprises bonding the substrate on the bottom silicon layer via an adhesive layer in between the substrate and the bottom silicon layer.

[0143] EEE 20 is the method according to EEE 13, wherein the processing of the fluidic channel comprises: [0144] laminating a dry-film-resist (DFR) on the top silicon layer, [0145] patterning the fluidic channel via lithography on the DFR, and [0146] hard-baking the patterned DFR.

V. Conclusion

[0147] The present disclosure is not to be limited in terms of the particular embodiments described in this application, which are intended as illustrations of various aspects. Many modifications and variations can be made without departing from its scope, as will be apparent to those skilled in the art. Functionally equivalent methods and apparatuses within the scope of the disclosure, in addition to those described herein, will be apparent to those skilled in the art from the foregoing descriptions. Such modifications and variations are intended to fall within the scope of the appended claims.

[0148] The above detailed description describes various features and operations of the disclosed systems, devices, and methods with reference to the accompanying figures. The example embodiments described herein and in the figures are not meant to be limiting. Other embodiments

can be utilized, and other changes can be made, without departing from the scope of the subject matter presented herein. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations.

[0149] With respect to any or all of the message flow diagrams, scenarios, and flow charts in the figures and as discussed herein, each step, block, operation, and/or communication can represent a processing of information and/or a transmission of information in accordance with example embodiments. Alternative embodiments are included within the scope of these example embodiments. In these alternative embodiments, for example, operations described as steps, blocks, transmissions, communications, requests, responses, and/or messages can be executed out of order from that shown or discussed, including substantially concurrently or in reverse order, depending on the functionality involved. Further, more or fewer blocks and/or operations can be used with any of the message flow diagrams, scenarios, and flow charts discussed herein, and these message flow diagrams, scenarios, and flow charts can be combined with one another, in part or in whole.

[0150] A step, block, or operation that represents a processing of information can correspond to circuitry that can be configured to perform the specific logical functions of a herein-described method or technique. Alternatively or additionally, a step or block that represents a processing of information can correspond to a module, a segment, or a portion of program code (including related data). The program code can include one or more instructions executable by a processor for implementing specific logical operations or actions in the method or technique. The program code and/or related data can be stored on any type of computer-readable medium such as a storage device including RAM, a disk drive, a solid state drive, or another storage medium.

[0151] The computer-readable medium can also include non-transitory computer-readable media such as computer-readable media that store data for short periods of time like register memory and processor cache. The computer-readable media can further include non-transitory computer-readable media that store program code and/or data for longer periods of time. Thus, the computer-readable media may include secondary or persistent long term storage, like ROM, optical or magnetic disks, solid state drives, compact-disc read only memory (CD-ROM), for example. The computer-readable media can also be any other volatile or non-volatile storage systems. A computer-readable medium can be considered a computer-readable storage medium, for example, or a tangible storage device.

[0152] Moreover, a step, block, or operation that represents one or more information transmissions can correspond to information transmissions between software and/or hardware modules in the same physical device. However, other information transmissions can be between software modules and/or hardware modules in different physical devices.

[0153] The particular arrangements shown in the figures should not be viewed as limiting. It should be understood that other embodiments can include more or less of each element shown in a given figure. Further, some of the illustrated elements can be combined or omitted. Yet further, an example embodiment can include elements that are not illustrated in the figures.

[0154] While various aspects and embodiments have been disclosed herein, other aspects and embodiments will be apparent to those skilled in the art. The various aspects and embodiments disclosed herein are for purpose of illustration and are not intended to be limiting, with the true scope being indicated by the following claims.

Claims

1. A micro-optomechanical sensor comprising: a surface comprising a cavity; a fluidic channel on the surface connected to the cavity, wherein the fluidic channel is configured to transport a fluid sample to the cavity; a cantilever on the surface being laterally extended inside the cavity, wherein at least one side of the cantilever is configured to be in contact with the fluid sample inside the

cavity; and an optical waveguide on the surface being extended to the cantilever, wherein the optical waveguide is configured to couple out a light signal from the cantilever in order to optically transduce a mechanical vibration of the cantilever.

2. The micro-optomechanical sensor according to claim 1, wherein the at least one side of the cantilever comprises a coating of at least one antibody and/or at least one antigen and/or at least one protein and/or at least one aptamer, corresponding to at least one biological cell in the fluid sample.

3. The micro-optomechanical sensor according to claim 2, wherein the mechanical vibration of the cantilever corresponds to a natural resonance frequency.

4. The micro-optomechanical sensor according to claim 2, wherein the mechanical vibration of the cantilever corresponds to a resonance frequency higher or lower than the natural resonance frequency due to an interaction of the at least one biological cell with the coating.

5. The micro-optomechanical sensor according to claim 1, wherein the surface further comprises an inlet connected to the fluidic channel, wherein the inlet is configured to receive the fluid sample and further to transport the fluid sample to the fluidic channel.

6. The micro-optomechanical sensor according to claim 1, wherein the cantilever is arranged on the surface in a non-overlapping manner with respect to the fluidic channel on the surface.

7. The micro-optomechanical sensor according to claim 1, wherein the optical waveguide is a dielectric waveguide.

8. The micro-optomechanical sensor according to claim 7, wherein the dielectric waveguide is a silicon-based waveguide.

9. The micro-optomechanical sensor according to claim 1, wherein the surface is a processed silicon-on-insulator (SOI) wafer.

10. A sensor system comprising: at least one micro-optomechanical sensor comprising: a surface comprising a cavity, a fluidic channel on the surface connected to the cavity, wherein the fluidic channel is configured to transport a fluid sample to the cavity, a cantilever on the surface being laterally extended inside the cavity, wherein at least one side of the cantilever is configured to be in contact with the fluid sample inside the cavity, and an optical waveguide on the surface being extended to the cantilever, wherein the optical waveguide is configured to couple out a light signal from the cantilever in order to optically transduce a mechanical vibration of the cantilever; at least one light source optically coupled to the optical waveguide of the at least one micro-optomechanical sensor, wherein the at least one light source is configured to generate the light signal for one or more optical waveguides of the at least one micro-optomechanical sensor; and a processing device optically coupled to the at least one micro-optomechanical sensor, wherein the processing device is configured to receive the light signal coupled out from each of the optical waveguides of the at least one micro-optomechanical sensor and further to optically transduce the mechanical vibration of one or more cantilevers of the at least one micro-optomechanical sensor.

11. The sensor system of claim 10, wherein the at least one side of the cantilever comprises a coating of at least one antibody and/or at least one antigen and/or at least one protein and/or at least one aptamer, corresponding to at least one biological cell in the fluid sample.

12. The sensor system according to claim 11, wherein the processing device is further configured to: process the light signal coupled out from the optical waveguide to detect a change in the resonance frequency of the mechanical vibration of the cantilever due to the interaction of the at least one biological cell with the coating, and identify at least one biological cell in the fluid sample based on the detected change in the resonance frequency.

13. A method for fabricating a micro-optomechanical sensor on a silicon-on-insulator (SOI) wafer comprising a dielectric layer in between a top silicon layer and a bottom silicon layer, the method comprising: defining a length of a cantilever and a length of a cavity on the SOI wafer; processing an optical waveguide on the top silicon layer, wherein the optical waveguide is extended over the length of the cantilever; processing a first etched groove at an edge of the length of the cantilever

within the length of the cavity by etching the top silicon layer, the dielectric layer, and partially the bottom silicon layer from the top silicon layer side; processing a second etched groove corresponding to the length of the cavity by etching the bottom silicon layer and the dielectric layer from the bottom silicon layer side, thereby forming the cavity in the bottom silicon layer and the cantilever on the top silicon layer laterally extended inside the cavity; bonding a substrate on the bottom silicon layer at least encompassing the cavity from the bottom silicon layer side; and processing a fluidic channel on the top silicon layer from the top silicon layer side.

14. The method according to claim 13, wherein the processing of the optical waveguide comprises: forming optical waveguide components on the top silicon layer from the top silicon layer side.

15. The method according to claim 13, wherein the processing of the optical waveguide comprises: bonding an additional layer on the top silicon layer and forming optical waveguide components on the additional layer from the top silicon layer side.

16. The method according to claim 13, wherein the processing of the first etched groove further comprises using the dielectric layer in between the top silicon layer and the bottom silicon layer as a etch stop layer.

17. The method according to claim 13, wherein the processing of the second etched groove further comprises using the dielectric layer in between the top silicon layer and the bottom silicon layer as a etch stop layer.

18. The method according to claim 13, wherein the bonding of the substrate on the bottom silicon layer comprises: bonding the substrate on the bottom silicon layer via oxide fusion between the substrate and the bottom silicon layer.

19. The method according to claim 13, wherein the bonding of the substrate on the bottom silicon layer comprises bonding the substrate on the bottom silicon layer via an adhesive layer in between the substrate and the bottom silicon layer.

20. The method according to claim 13, wherein the processing of the fluidic channel comprises: laminating a dry-film-resist (DFR) on the top silicon layer, patterning the fluidic channel via lithography on the DFR, and hard-baking the patterned DFR.
